

Title

Sustaining Open-Access Journals Through AI-Driven Editorial Automation

Abstract

Open-access (OA) journals have become a central pillar of contemporary scholarly communication by enabling unrestricted access to scientific knowledge. However, sustaining open-access publishing infrastructures remains a significant challenge due to financial constraints, increasing submission volumes, and the labor-intensive nature of editorial workflows. Artificial intelligence (AI) technologies offer promising opportunities to automate aspects of journal management, including manuscript screening, reviewer selection, plagiarism detection, and editorial decision support. This paper examines the role of AI-driven editorial automation in supporting the sustainability of open-access journals. By reviewing current developments in digital publishing technologies and editorial management systems, the study explores how AI can improve efficiency, reduce operational costs, and enhance the scalability of open-access publishing models. The analysis suggests that while AI-driven automation can significantly streamline editorial workflows, careful governance, transparency, and human oversight are essential to maintain fairness, research integrity, and trust in the scholarly publishing process.

Introduction

Open-access publishing has transformed the landscape of scholarly communication by making research outputs freely accessible to readers worldwide. Unlike traditional subscription-based models, open-access journals allow unrestricted access to scientific articles, thereby promoting knowledge dissemination and supporting global research collaboration (Suber, 2012). Over the past two decades, open-access publishing has expanded rapidly, supported by institutional mandates, funding agency requirements, and initiatives promoting open science.

Despite its benefits, sustaining open-access journals presents substantial challenges. Many journals operate with limited financial resources and rely heavily on volunteer editorial labor. Increasing submission volumes further strain editorial teams responsible for managing peer review, evaluating manuscripts, and coordinating publication workflows (Tennant et al., 2017).

Artificial intelligence (AI) technologies have recently emerged as potential tools for improving the efficiency of editorial processes. Machine learning and natural language processing systems can analyze manuscripts, detect plagiarism, recommend reviewers, and assist editors in decision-making tasks (Kasneci et al., 2023). These capabilities offer opportunities to reduce administrative burdens and enable editorial teams to manage large numbers of submissions more effectively.

However, the adoption of AI-driven editorial automation also raises important concerns regarding transparency, bias, and accountability in scholarly publishing. Automated systems must be carefully designed to ensure that editorial decisions remain fair and that algorithmic processes do not undermine academic integrity.

This paper explores the role of AI-driven editorial automation in sustaining open-access journals. By examining emerging technologies and their potential applications in editorial workflows, the study highlights opportunities and challenges associated with integrating AI into open-access publishing infrastructures.

Literature Review

Open-access publishing has gained significant momentum since the early 2000s, driven by initiatives such as the Budapest Open Access Initiative and institutional open-access mandates. Scholars emphasize that open-access models improve research visibility, increase citation rates, and enhance global access to scientific knowledge (Suber, 2012).

However, open-access publishing models face financial sustainability challenges. Article processing charges (APCs), institutional funding, and volunteer labor often support editorial operations. Managing editorial workflows—including manuscript submission, reviewer coordination, and editorial decision-making—requires substantial administrative effort (Tennant et al., 2017).

Digital publishing technologies have helped streamline some aspects of journal management. Editorial management systems enable automated submission tracking, communication with reviewers, and publication scheduling. Nevertheless, these systems still require extensive human involvement in evaluating manuscripts and coordinating peer review.

Recent research explores how artificial intelligence can further enhance editorial workflows. Natural language processing techniques can analyze manuscript content to identify subject areas, detect plagiarism, and flag potential ethical concerns (Kasneci et al., 2023). Machine learning models can also recommend potential reviewers by analyzing citation networks and author expertise.

Another area of development involves automated manuscript screening tools. These systems evaluate submissions for compliance with journal guidelines, formatting standards, and ethical policies. Automated screening can reduce the workload for editorial staff by filtering out unsuitable submissions before peer review.

Despite these advancements, scholars emphasize the need for careful oversight when implementing AI in editorial processes. Automated systems may reproduce biases present in training datasets or produce inaccurate recommendations. Ensuring transparency and maintaining human oversight are therefore critical considerations for responsible implementation of AI-driven editorial automation.

Discussion

AI-driven editorial automation has the potential to address several challenges associated with sustaining open-access journals.

Manuscript Screening and Quality Control

One of the most promising applications of AI in editorial workflows is automated manuscript screening. Machine learning models can analyze submitted manuscripts to detect plagiarism, identify incomplete references, and assess compliance with journal formatting guidelines.

Such systems enable editors to quickly identify manuscripts that require further review or that fail to meet basic submission criteria. Automated screening reduces the time required for preliminary editorial assessment and allows editorial teams to focus on evaluating high-quality submissions.

Reviewer Recommendation Systems

Identifying appropriate peer reviewers is often one of the most time-consuming tasks in journal management. AI-based reviewer recommendation systems analyze citation networks, publication histories, and subject classifications to identify potential reviewers with relevant expertise.

These systems can significantly accelerate the reviewer selection process while improving the alignment between manuscript topics and reviewer expertise.

Editorial Decision Support

AI tools can also assist editors in evaluating manuscripts by providing structured summaries of research contributions, highlighting potential methodological concerns, and identifying related publications. Such tools can help editors navigate complex submissions more efficiently.

However, AI-generated analyses should be treated as decision-support mechanisms rather than substitutes for editorial judgment. Editors remain responsible for evaluating the quality and significance of submitted research.

Ethical and Governance Considerations

While AI-driven automation offers substantial efficiency gains, ethical considerations must guide its implementation. Algorithmic bias in reviewer recommendations or manuscript evaluations could inadvertently disadvantage certain authors or research topics.

Transparency in AI systems is therefore essential. Journals should clearly communicate how automated tools are used in editorial workflows and ensure that human oversight remains central to decision-making processes.

Additionally, editorial teams must regularly evaluate the performance of AI systems to ensure that they produce reliable and unbiased results.

Conclusion

Sustaining open-access journals requires innovative approaches to managing editorial workflows and reducing operational costs. AI-driven editorial automation offers significant potential to streamline journal management processes by automating tasks such as manuscript screening, reviewer selection, and content analysis.

These technologies can enhance the scalability and efficiency of open-access publishing while allowing editorial teams to focus on the intellectual evaluation of research contributions. However, successful implementation requires careful attention to transparency, fairness, and ethical governance.

Human oversight remains essential to ensure that AI tools support rather than replace editorial judgment. By combining automated systems with responsible governance frameworks, open-access journals can harness the benefits of AI-driven automation while maintaining the integrity and trustworthiness of the scholarly publishing process.

Future research should explore the long-term impacts of AI-assisted editorial workflows on peer review quality, publication efficiency, and the sustainability of open-access publishing models.

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